

Introduction to Health Informatics

Learning Objectives:

- 1. To understand the concept of Health Informatics**
- 2. To understand the elements of Health Informatics**
- 3. To understand the Health Informatics Context**
- 4. Define the concept of an information system and its characteristics**
- 5. Describe the different types of information systems**
- 6. Describe various types of technologies that support health care information systems**
- 7. Examine the challenges presented by emerging trends in information technology, social media, and global communications**
- 8. Discuss the advantages and disadvantages of using the Internet as a platform for health care applications**

The (e)ATM of Healthcare



HIS

- A health information system (HIS) is any form of structured repository of data, information, or knowledge that can be used to support health care delivery or to promote health development.
- HIS should be part of the health technology assessment framework. Data is an essential element of HIS and it is usually an expensive resource. Investments in HIS should look into its opportunity costs and the efficacy/effectiveness of its potential contribution to health development

Why Health Informatics?

- Health Informatics provides information to make decisions
- Better information leads to better decisions
- Health care, management, planning and policy all need good information

good information is needed to make
decisions on

- Health care,
- health management,
- health policy and
- health planning

Health Informatics

- *The intersection of information science, computer science, and health care.*
- *It deals with the resources, devices and methods required to optimize the acquisition, storage, retrieval and use of information in health.*

Health informatics tools include

Tools

- *computers*
- *clinical guidelines,*
- *formal medical terminologies,*
- *Information and communication systems.*

Definition of health information system

- An automated system that uses computer hardware and software to record, manipulate, store, recover, and disseminate data.
- that is, a system that receives and processes input and provides output.

- Inputs
 - Order for a drug
- Data
 - Prescribed drug
- Process
 - Verifies drug is correct for the order
- Outputs

Categories of IS

With regards to information systems, there are three major categories. They are

- Transaction Processing System (TPS),
- Management Information System (MIS),
and
- Decision Support System (DSS)

A transaction processing system (TPS)

- A transaction processing system (TPS) processes information in order to complete a transaction.
- Two examples are the Admit, discharge, transfer (ADT) and patient billing systems.

An admission, discharge, and transfer system(ADR)

- This is the name given to software systems used in health care facilities that register and track patients from admission through discharge including transfers; usually interfaced with other systems used throughout a facility such as an electronic health record or lab information system”

Management Information System (MIS)

Helps managers plan, organize, and evaluate their department

- Examples:
 - Laboratory department
 - Emergency department

Emergency care INFORMATION SYSTEM

- Dashboard systems
- Communication systems
- Clinical decision support
- Documentation systems
- Computerized training and information retrieval
- Systems to facilitate public health surveillance

Emergency care INFORMATION SYSTEM

- “dashboard systems that track and coordinate patient flow,
- communications systems that enable ED physicians to link to patients’ records or providers,
- clinical decision support programs that improve decision making,
- documentation systems for collecting and storing patient data,
- computerized training and information retrieval, and systems to facilitate public health surveillance

Decision support system (DSS)

- Information system
 - Gathers data from a variety of sources
 - Assists in providing structure to the data
 - Facilitates and improves the outcome of business and organizational decision-making activities
- Example:
 - Clinical DSS

Health Care Information System (HCIS)

- Information system used within a health care organization
 - Facilitates communication
 - Integrates information
 - Documents health care interventions
 - Performs record keeping
 - Otherwise support the functions of the organization

HCIS components

- 1) Patient management and billing
- 2) Department management
- 3) Care delivery and clinical documentation
- 4) Clinical decision support
- 5) Financial and resource management

Patient Management and Billing

- Systems that support the management of the patient
 - Example
 - Patient identification
 - Supporting technology
 - Master patient index
 - Patient-identifying directory
 - Links to the patient record
 - Facilitates patient identification
 - Assists in maintaining a longitudinal patient record

MoH Health Informatics

- provides leadership in the application of information and computer science and technology to public health practice, research, and learning.
 - Electronic health record support the public health functions
 - Use of health care records, population census and other public health data in supporting public health systems and analyses
 - Basic capabilities that support public health practice such as statistical and health surveillance
 - Public Health decision support

eHealth

- **eHealth** is a broad term for healthcare practice which is supported by electronic processes and communication.
- The term can encompass a range of services that are at the edge of medicine/healthcare and information technology.

mHealth

- Mobile Health
- Mobile technologies such as mobile phones collect and access health information.

Informatics $\neq$ IT

- Information Technology is not Informatics
- Information technology is hardware & software.
 - IT is to nouns, as informatics is to verbs.
 - Informatics helps IT 'work appropriately.'

Key Elements of Informatics

- Acquisition
- Storage
- Communication
- Manipulation
- Display

Health Information Systems Context



Community

Data Collection

Facility

Data Collection
and Indicators

District

Data
Aggregation
and Analysis

National

Data
Aggregation
and Analysis

International

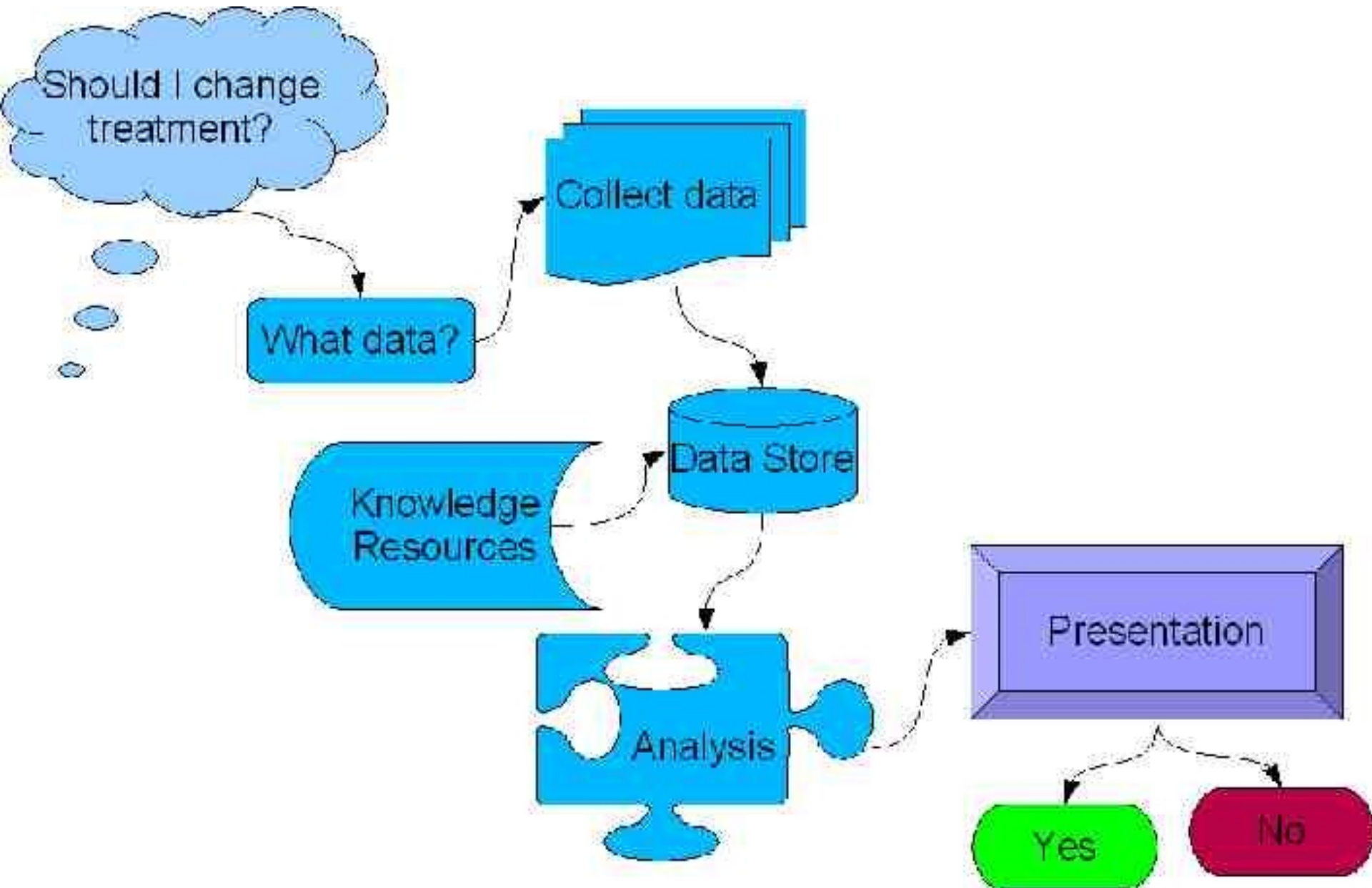
Data
Aggregation
and Analysis

Monitoring, Evaluation, and Management

Health Informatics Principles

- Use drives data
- Interoperability using open standards
- Incremental development and strengthening of systems
- Enterprise Architecture approach
- Collaborative Communities

Decisions Determine Data



Encourage open systems



Make vs. Buy... Or Modify

- Buy Software
 - May not be an exact fit to your needs
- Build Software
 - Long expensive process not guaranteed to succeed.
- Modify
 - Start with open source software that you can modify
 - Modified software to meet your exact requirements
 - Everyone benefits from your investment in the software

Collaborative communities of practice

Collaboration Life-Cycle





TeleConsultations previously Telemedicine Services

The *eight* Drivers Forcing Change TO TELEMEDICINE

- Obligatory search for cost containment
- Changing demographics
- Peripheralisation of health care delivery
- Changing disease patterns
- Impact of information technology including IT support for clinical decisions - *Telemedicine & eHealth*
- More informed and expectant patients C-Health
- The well-being factor – responsibility shift into patients hands - The Wellness Paradigm
- Knowledge Management – changing world

Examples of Telemedicine/ E-Health Applications

- Diagnosis and treatment of patients at distant locations
- Monitoring of patients with chronic illness/disease
- Automated message reminders (e.g., medications, appointments)
- Physician consultations
- Patient portals of EHR systems



Examples of Telemedicine/ Health Applications

E-



Potential Advantages of Technology

- Eliminates the need for patients and health care professionals to travel
- Permits more flexible appointment scheduling
- Permits easier access to a wider array of information and services especially for rural and underserved patients
- Permits increased access to specialists
- Permits asynchronous communication
- Patient empowerment (e.g., well-informed questions; information on treatment options)

Potential Disadvantages of Technology

- Proliferation of incorrect or inappropriate information
- Proliferation of too much information
- Potential disruptions in physician/patient relationships
- Privacy issues
- Technology breakdowns/failures
- Promotes false sense of security
- Potential for increasing health care disparities

Critical Question

Will older people be able to access and successfully adapt to and use telemedicine and e-health tools?

Requirements of Meaningful Access to Health IT

- Awareness of technology
- Access to technology and technical support
- Requisite knowledge and skills to use the technology
- Able to comprehend and integrate the information and use the information appropriately

Barriers to Use of Healthcare Technologies

Limited knowledge of potential uses/benefits of technology

Cost

Rapid developments and changes in technical systems

Interphace issues

Use of most telemedicine and E-health tools require at least adequate levels of health literacy

Barriers to Use of Healthcare Technologies

- Older adults especially minority have lower health literacy than other segments of the populations
- Human-technology interaction involves information processing and places demands on memory and other cognitive abilities
- Age is associated with declines in some aspects of cognition

The Internet enables patients to make more informed decisions

- Majority (59%) of physicians felt the Internet enables patients to make more informed decisions regarding health choices
 - Helps patient understand their condition and their medications
 - Helps patient in asking questions during the visit
 - Allows patient to find out about more options than what was discussed by their provider
 - Allows patient to be their own advocate

How can patients better prepare for office visits?

- If using the Internet to get information, use major sites and stay away from commercial promotions; check the source and weigh the information accordingly
- To improve communication, patients should not think that they have all of the answers based upon what they read online and be open to discussion with their physician

How can patients better prepare for office visits?

- “...you can’t expect to do medical school and residency training online in front of your computer in an evening before you go in to see your doctor... [use] what information you gather in order to spark discussion and [gain] information but [don’t come] to too many conclusions on something that is probably quite complicated.”

Electronic Medical Records

- Examining the ability of older adults to use patient portals of EMRs to perform routine tasks and identify usability problems
- Also gathering information on issues related to privacy and trust
- Focus on lower SES populations with low health literacy